This procedure is applicable to the following models:

Machine Model	Server Option		
	S1	S2	S3
V810 / V810i STD			
V810 / V810i XXL			
V810 / V810i S2EX		√	√
V810 / V810i Mini			
V810 / V810i XLT / XLTM / XLL		√	√
V810 / V810i XLW			
V810 / V810i S3		√	√

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	30 APR 2024	First Release	Yeap Leong Wei
01	30 OCT 2024	Content Update	Yeap Leong Wei

Content

Scope

This procedure provides guidelines for the removal and replacing the X-Axis Quantic scale.

Tool List

No	Part No	Description	Qty	UOM	Images
1	-	Ball Point Metric Allen Wrench Set	1	SET	
2	-	Metric T-Handle Wrench (M5 size)	1	PC	
3	-	Camera Tool	1	PC	
4	-	Temper Proof Screw Driver Set	1	Set	
5	-	Cutter	1	PC	

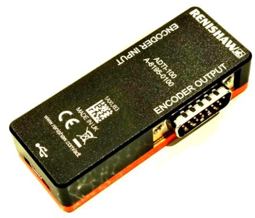
6	4XASPP-A440	Renishaw Quantic Encoder Alignment Kit	1	Set	 +  + 

Table 1: Tool List

Origin X-Axis Quantic Scale Removal Procedure

1. **Board Unloading:** Prior to commencing the removal procedure, ensure that any boards within the machine are safely unloaded. Follow the manufacturer's guidelines or equipment manual for proper board removal procedures.
2. **Shutdown Sequence:** Close V810 GUI and initiate the shutdown sequence for the Super Micro PC by following the manufacturer's recommended procedure. Ensure all data is saved and any necessary backups are performed before proceeding.
3. **Deactivation of Equipment:** At the operator console panel, press the "X-Ray Off" button to deactivate any X-ray equipment. Turn the key to the "Off" position to ensure complete deactivation of the system and associated components.
4. **Power Down Procedure:** Safely power down the SSCA main power source by following the manufacturer's recommended shutdown procedure. Ensure all power indicators are off before proceeding with the removal process.
5. **Access Door Opening:** Open all front access doors carefully to provide sufficient access to the internal components of the machine. Ensure doors are fully opened and secured to prevent accidental closure during the removal process.
6. **Left Outer Barrier Removal:** Remove the left outer barrier by unscrewing the door screws using the appropriate tools. Store the screws in a secure location for reassembly after the removal process is complete.
   a) For S2EX & XLT

Figure 1:Left Outer Barrier (S2EX & XLT)

b) For S3

Figure 2:Left Outer Barrier (S3)

7. **Rear Access Door Opening:** Open the rear access door carefully, ensuring there are no cables or obstructions that may hinder the removal process. Take caution to prevent strain on any cables or connectors located near the access door.
8. **Loosen the Screws for Spring Clamp:** Loosen the screws for the spring clamp with M3 Allen key.

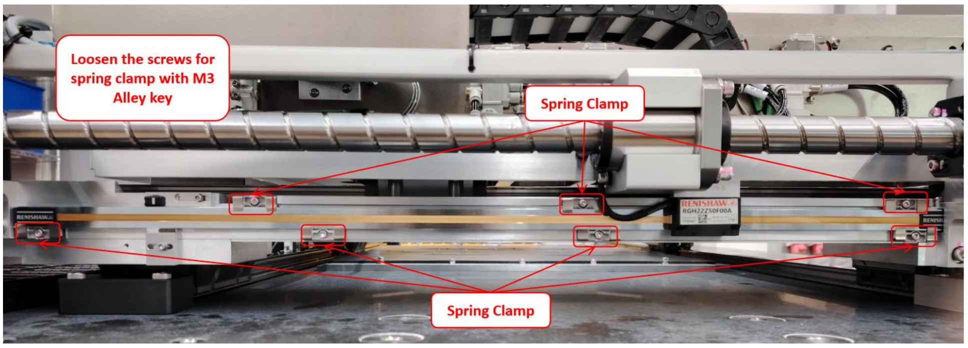


Figure 3:X-Axis Quantic Scale

9. **Removal of Steel Bar with Clamps (Top Side):** Unfasten the screws for steel bar with clamps. Carefully remove and place it on the clean and flat surface.

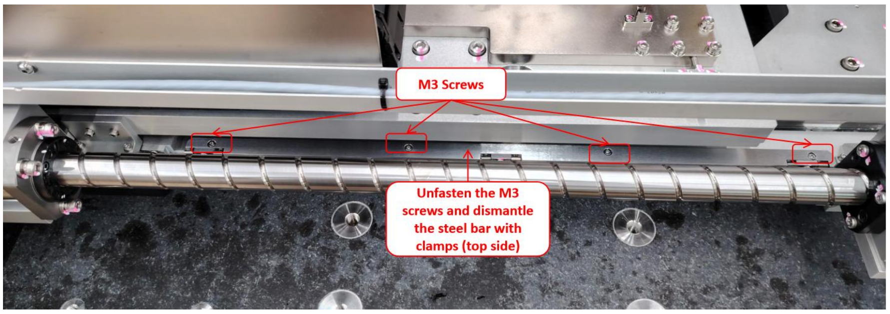


Figure 4:X-Axis Cover Plates

10. **Removal of X-Axis Bracket:** Gently remove the X-Axis encoder bracket and allocate it in a visible position.

Figure 5:X-Axis Encoder Bracket

11. **Removal of X-Axis Quantic Scale:** Carefully remove the X-Axis Quantic scale. Take caution to place the scale in a flat surface to prevent any incident.

Figure 6:Quantic Scale

New X-Axis Quantic Scale Replacement

12. **Spare Part Preparation:** Refer to the table below to gather and prepare the new spare part required for the replacement process.

Part number	Part Description	QTY	UOM	illustration
4XASPP-A430	S2EX X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4XASPP-A432	XLT X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	
4XASPP-A434	S3 X-AXIS QUANTIC ENCODER SCALE ASSEMBLY KIT	1	PC	

Table 2: X-Axis Quantic Scale List for S2EX, XLT and S3

13. **X-Axis Quantic Scale Assembly:** Assemble the Quantic scale and allow it to rest on the spring clamps. Be cautions for any obstacle and not damaging the Quantic scale.

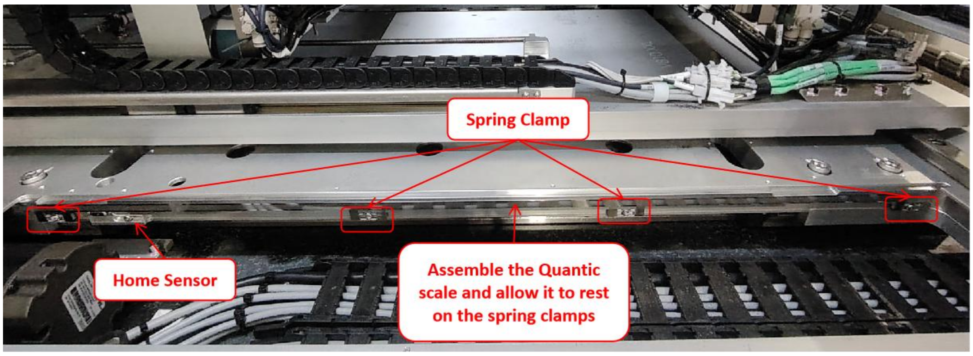


Figure 7:X-Axis Quantic Scale (Sample Image)

14. **Steel Bar with Clamps (Top side) Assembly:** Assemble the steel bar with clamps that guiding the top side of the Quantic scale. Tighten the M3 screws after align the steel bar in position.

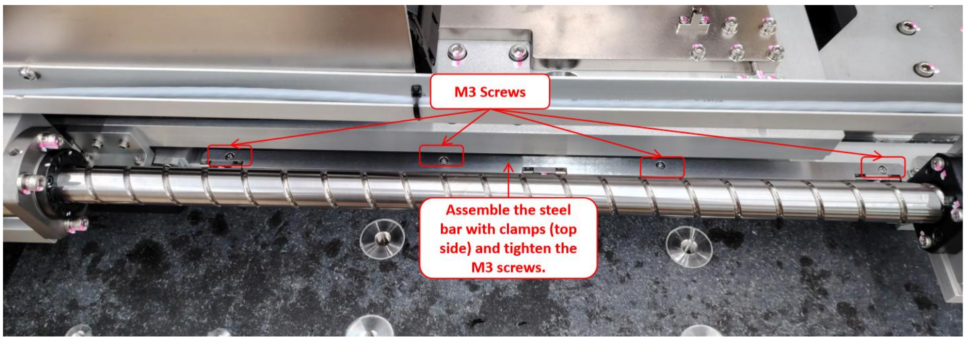


Figure 8:X-Axis Encoder Cover Plate and Limit Sensor (Sample Image)

15. **Home and Limit Sensors Assembly:** Align the home sensor and tighten the screw in the center position of the elongate. Then, assemble the limit sensor and tighten the M3 screw.

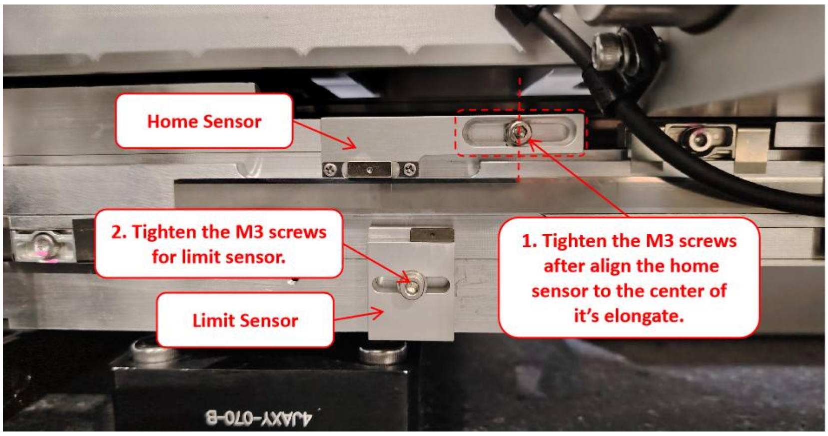


Figure 9:X-Axis Home and Limit Sensors Alignment

16. **X-Axis Encoder Bracket Assembly:** Assemble the X-Axis encoder bracket and tighten the 2*M3 screws. Be cautious not to damage the read head encoder during installation process.

Figure 10:X-Axis Encoder with Bracket

Reference for Alignment Procedure

17. **X-Axis Encoder Read Head Alignment**: Refer to "**AXI V810 Quantic Encoder Read Head Alignment Procedure**" to verify the encoder status and perform necessary alignment if needed.

    Note: Crucial to ensure the signal strength for encoder read head along the x-axis stroke.

18. **Home and Limit Sensors Alignment**: After validating the encoder head performance, refer to **AXI V810 Quantic Home and Limit Sensors Alignment Procedure** and perform necessary alignment and adjustment.

    Note: It's a must perform procedure whenever there is any adjustment or replacement of Quantic scale.